

Basic Level

Qn.1.Problem Statement:

"Enter N numbers. One number is missing. Find it."

Input 1:

Enter the value of N: 5

Enter 4 space-separated numbers: 1 2 3 5

Output: 4

Input 2:

Enter the value of N: 7

Enter 6 space-separated numbers: 1 2 4 5 6 7

Output: 3

Qn.2. Problem Statement:

"Enter a list of numbers. One number appears the most. Find it."

Input 1:

Enter space-separated numbers: 1 3 2 3 4 3 5 2 3

Output: 3

Input 2:

Enter space-separated numbers: 7 8 9 7 6 7 5 4 7 6 6 6

Output: 7

Intermediate Level

Problem Statement:

1. "Enter numbers and a target sum. Find all unique pairs that add up to it."

Input 1:

Enter space-separated numbers: 1 3 2 2 4 5

Enter the target sum: 5

Output:

Output: [(1, 4), (3, 2)]

Input 2:

Enter space-separated numbers: 10 15 3 7 8 5 2

Enter the target sum: 10

Output:

Output: [(3, 7), (8, 2)]

Advanced Level**Problem Statement:**

You are given an array of numbers. A cluster is defined as a sequence of consecutive numbers. Your task is to find all such clusters in the given array.

Example Input & Output:

Input 1: Enter space-separated numbers: 1 2 3 7 8 10 11 12 15

Output : [[1, 2, 3], [7, 8], [10, 11, 12], [15]]

Input 2 : Enter space-separated numbers: 5 6 7 20 25 26 27 28

Output : [[5, 6, 7], [20], [25, 26, 27, 28]]